Anomaly Detection for Healthcare Applications

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***Özetçe*—Bu çalışma kapsamında yapılan araştırmalar sonucunda, sağlık uygulamalarında anomali tespiti problemini ele almakta ve biyomedikal zaman serisi verilerindeki anormal örüntülerin belirlenmesine odaklanmaktadır. Sıralı sağlık verilerinin analizinde karşılaşılan başlıca zorluklar tartışılmakta ve normal ile anormal örüntüleri ayırt etmek amacıyla derin öğrenme tabanlı yaklaşımlar incelenmektedir. Önerilen yöntemlerin fizyolojik sinyallerin analizine nasıl katkı sağlayabileceğini ve anormal durumların erken tespitini nasıl destekleyebileceğini göstermek amacıyla, temsili bir uygulama olarak EKG sinyalleri üzerinde 1D-CNN ve LSTM mimarileri değerlendirilmiştir.**

***Anahtar Kelimeler*—anomali tespiti, sağlık uygulamaları, EKG sinyalleri, zaman serisi, derin öğrenme**

***Abstract*—This study addresses the problem of anomaly detection in healthcare applications, focusing on the identification of abnormal patterns in biomedical time-series data. We discuss the principal challenges associated with analyzing sequential healthcare information and examine deep learning-based approaches for distinguishing normal from anomalous patterns. As a representative application, ECG signals are used to evaluate 1D-CNN and LSTM architectures, demonstrating how such methods can support the analysis of physiological signals and contribute to the early identification of abnormal cardiac conditions.**

Code: https://github.com/cerenyildiiz/ecg5000-anomaly-detection-time-series

***Keywords*—anomaly detection, healthcare applications, ECG signals, time series, deep learning**

I. Introduction

Anomaly detection is the task of finding patterns in data that do not match what is expected. In many real-world settings, these unusual patterns are exactly what matters most: a sudden spike in machine temperature, a fraudulent transaction, or, in healthcare, an irregular heartbeat. A human observer can often catch these irregularities, but doing so reliably across large amounts of data is slow and prone to error.

Machine learning methods have been applied across many domains for tasks such as early diagnosis, anomaly detection, failure prediction, and predictive maintenance (Sarker, 2021). In healthcare, these methods are especially useful, as they can uncover patterns in patient data that may not be obvious to a human observer. This study focuses on ECG signals, which record the heart's electrical activity over time. A healthy heart produces a fairly regular waveform, so deviations from this pattern can point to an underlying cardiac condition. We explore how deep learning architectures such as 1D convolutional neural networks (1D-CNN) and LSTM networks can be used to detect these abnormal heartbeats directly from raw signal data, without relying on manual feature engineering.

The overall workflow of the study is summarized in Fig. 1. The ECG5000 data are first explored, and the heartbeat classes are then modelled with a 1D-CNN and an LSTM autoencoder. The models are evaluated with standard classification metrics, and the model decisions are finally interpreted with SHAP.

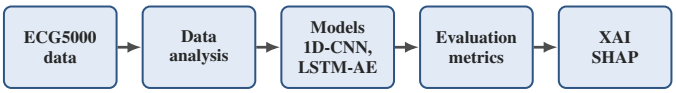


Fig. 1. Overall workflow of the study.

II. Related Work

Anomaly detection in ECG signals has been studied with a wide range of deep learning models, and the ECG5000 dataset is one of the most commonly used benchmarks for this task. Table I summarizes representative studies and the performance reported by their authors.

Adegoke-Elijah et al. propose an LSTM autoencoder for ECG anomaly detection and evaluate it on ECG5000, reporting an accuracy of 99.23% and an AUC of 98.00% (Adegoke-Elijah et al., 2025). They argue that unsupervised methods such as K-means clustering and Hierarchical Temporal Memory (HTM) often perform poorly because they depend on pre-defined features. Nayana and Vinay also use an LSTM autoencoder on ECG5000, but combine it with SHAP so that the temporal segments that influence an anomaly prediction can be visualized; their framework reaches an accuracy of 97.6% and an F1-score of 96.7%, and it is compared with SVM and Logistic Regression baselines (Nayana and Vinay, 2026).

Attention-based and generative models have also been explored. Alamr and Artoli introduce an unsupervised model that consists of an embedding layer and a standard transformer encoder, and detect anomalies from the loss between the real and the predicted ECG sequences; the model is validated on ECG5000 and MIT-BIH Arrhythmia (Alamr and Artoli, 2023). Shi and Jiang propose LAVAE-Net, which combines a variational autoencoder with an LSTM encoder enhanced by an attention mechanism and a GRU decoder. The model is trained exclusively on normal heartbeats, so that no labelled anomalies are required, and reaches an F1-score of 99.4% on ECG5000 (Shi and Jiang, 2026). Wu et al. replace the encoder of the autoencoder with adaptive convolution kernels and Toeplitz matrices designed around the known shape of

ECG time series, which improves accuracy and also reduces the runtime (Wu et al., 2025).

LSTM-based anomaly detection is not limited to ECG data. Paul and Mishra use an LSTM autoencoder to detect insider threats from the daily activity sequences of employees, where a threat is flagged through a significant loss from the normal routine (Paul and Mishra, 2020). Aggarwal applies LSTM-based anomaly detection to United States export and import time series between 1992 and 2022 and explains the role of the input, forget and output gates (Aggarwal, 2023). These studies show that reconstruction- and prediction-based sequence models are applicable across very different domains.

Table I
REPRESENTATIVE STUDIES ON ECG5000-BASED ANOMALY DETECTION (PERFORMANCE AS REPORTED BY THE ORIGINAL AUTHORS; THE RESULTS ARE NOT DIRECTLY COMPARABLE BECAUSE THE EVALUATION PROTOCOLS DIFFER).

Study	Model	Dataset	Reported result
Adegoke-Elijah et al. (2025)	LSTM autoencoder	ECG5000	Acc. 99.23%, AUC 98.00%
Nayana and Vinay (2026)	LSTM autoencoder + SHAP	ECG5000	Acc. 97.6%, F1 96.7%
Alamr and Artoli (2023)	Transformer encoder	ECG5000, MIT-BIH	Acc. 99% (ECG5000), 89.5% (MIT-BIH)
Shi and Jiang (2026)	LSTM-attention VAE	ECG5000	F1 99.4%
Wu et al. (2025)	Toeplitz convolution classifier	ECG5000	Acc. above 99.6%

Most of these studies focus on detection performance, while only a few address the interpretability of the predictions. In this study, a 1D-CNN is first examined as a baseline model for heartbeat classification, an LSTM autoencoder is considered for reconstruction-based anomaly detection, and SHAP is planned for explaining the model decisions over the time dimension.

III. CONVOLUTIONAL NEURAL NETWORK (CNN)

Convolutional Neural Networks (CNNs) are among the most widely used deep learning architectures.

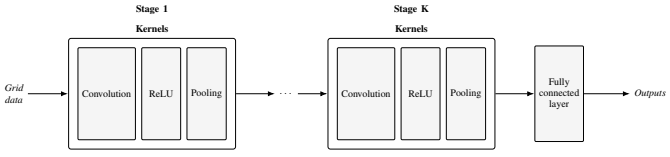


Fig. 2. General CNN architecture (Gharehbaghi, 2023).

A. 1D Convolutional Neural Network (1D-CNN)

As shown in Fig. 2, a single kernel in Stage 1 performs a one-dimensional convolution over the input data. This type of network is well suited to processing one-dimensional data such as time series and signal data, which is why this study begins with the introduction of the 1D CNN architecture.

B. Mathematical Formulation

A convolutional layer slides a set of learnable kernels over the input signal (Gharehbaghi, 2023). For an input signal x of length L with C channels, the output of the j-th filter at position t is

$$y_j[t] = b_j + \sum_{c=1}^{C} \sum_{k=0}^{K-1} w_{j,c}[k]\; x_c[s\,t + k - p], \qquad (1)$$

where K is the kernel size, s the stride, p the padding, $w_{j,c}$ the kernel weights and b_j the bias. The length of the output is

$$L_{\text{out}} = \left\lfloor \frac{L + 2p - K}{s} \right\rfloor + 1. \qquad (2)$$

For the input length $L = 140$ and the settings $K = 8$, $s = 2$ and $p = 3$ used in this study, this gives $L_{\text{out}} = \lfloor (140 + 6 - 8)/2 \rfloor + 1 = 70$, which matches the feature map size of 2×70 in Fig. 5.

The convolution output is passed through the ReLU activation, $\mathrm{ReLU}(z) = \max(0, z)$, which introduces non-linearity. Max pooling with a kernel size of 2 then keeps the larger of every two neighbouring values, which halves the length of the feature map from 70 to 35. After flattening, the resulting 70 features are passed through two fully connected layers, and the output layer converts the final scores z_k into class probabilities with the softmax function,

$$\hat{y}_k = \frac{e^{z_k}}{\sum_i e^{z_i}}. \qquad (3)$$

The network is trained by minimizing the mean squared error between the predictions and the targets over N training samples,

$$\mathcal{L} = \frac{1}{N} \sum_{n=1}^{N} \|\hat{\mathbf{y}}_n - \mathbf{y}_n\|_2^2 . \qquad (4)$$

IV. ECG5000 DATASET

The ECG5000 dataset contains 5,000 ECG heartbeat samples. Each sample consists of 140 numerical features representing the ECG signal and 1 target variable (class). There are no missing values in the dataset. More details about the dataset can be found on the OpenML website (Ferreira, 2022).

A. ECG5000 Classes

The ECG5000 dataset includes five different heartbeat classes. Since this study focuses on anomaly detection, these classes are grouped into normal and anomalous heartbeats as follows:

1) **Normal Beat (N)**
2) **R-on-T Premature Ventricular Contraction (R-on-T PVC)**
3) **Premature Ventricular Contraction (PVC)**
4) **Supraventricular Premature or Ectopic Beat (SP)**
5) **Unclassified Beat (UB)**

The class distribution of the dataset is shown in Fig. 3. The dataset is highly imbalanced: Normal beats (2,919 samples)

and R-on-T PVC beats (1,767 samples) make up the vast majority of the data, while PVC (96), SP (194) and UB (24) beats are rare.

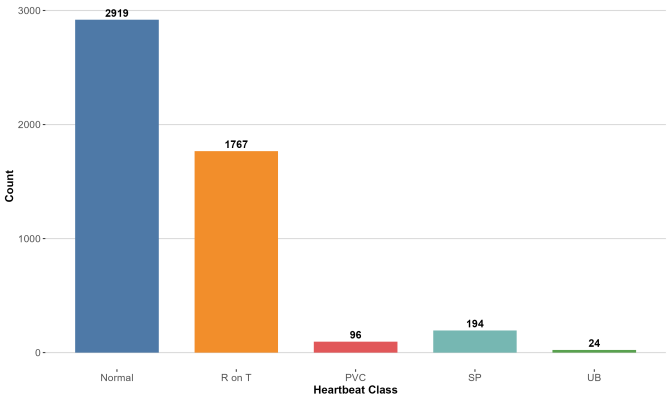


Fig. 3. Class distribution of the ECG5000 dataset.

Fig. 4 shows the average ECG waveform of each class over the 140 time points, together with a shaded band that indicates the variability of the signals around the mean. The Normal class has a clear and distinctive shape: after the initial rise, the signal shows a pronounced peak of about 1.5 around time point 105, followed by a dip and a second smaller rise towards the end of the heartbeat. Its band is also relatively narrow, which means that normal heartbeats are quite consistent with each other.

In contrast, the anomalous classes lose this characteristic structure. The mean waveforms of R-on-T PVC and PVC are much smoother, with a broad plateau instead of a sharp peak, and both end with a steep drop. The SP class follows a similar overall trend but with a slightly different amplitude profile. The UB class has the widest band throughout the heartbeat, showing the highest variability among its samples; this is probably related to the fact that it is both a mixed "unclassified" group and the smallest class (24 samples). These visible differences between the normal mean waveform and the anomalous ones suggest that the classes can be distinguished from the raw signal, which motivates the use of 1D-CNN and LSTM models in this study.

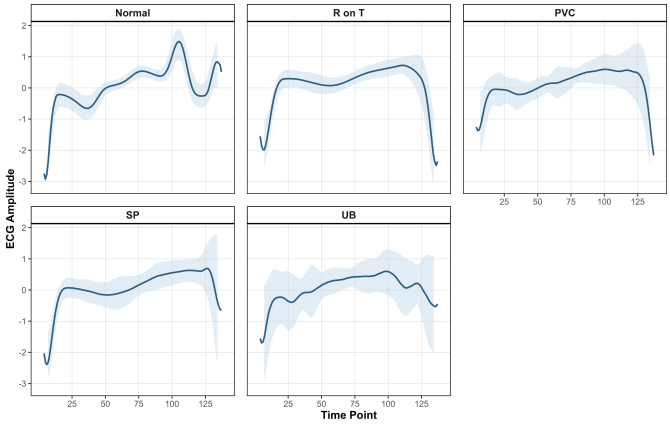


Fig. 4. Mean ECG waveform of each heartbeat class in the ECG5000 dataset. The shaded areas show the variability around the mean.

B. Binary Labelling and Class Imbalance

Since this study focuses on anomaly detection, the five classes are grouped into two categories: the Normal class (class 1) is treated as normal, and the R-on-T PVC, PVC, SP and UB classes (classes 2–5) are treated together as anomalous. According to the class counts in Fig. 3, this gives 2,919 normal beats (58.4%) and 2,081 anomalous beats (41.6%). The binary problem is therefore only moderately imbalanced, whereas the imbalance among the four anomalous classes themselves is much stronger: the 24 UB samples correspond to less than 1% of the data. This is important for multi-class evaluation, because a model that ignores the rare classes could still achieve a high overall accuracy.

The original dataset is distributed as separate training and test files, which are combined in this study into a single set of 5,000 heartbeats. Each heartbeat is a fixed-length sequence of 140 amplitude values, so no additional segmentation is required, and the dataset has no missing values.

V. 1D-CNN Model Training

Following the 1D-CNN architecture introduced earlier, a single-layer 1D-CNN is trained on the 140-point heartbeat signals. Each heartbeat is treated as a sequence of 140 amplitude values, where the amplitude represents the electrical potential of the heart measured at each time point. The model is trained with the Adam optimizer using the mean squared error (MSE) loss function. The hyperparameters used in the experiments are summarized in Table II, and the architecture of the model is shown in Fig. 5.

Table II
HYPERPARAMETERS OF THE 1D-CNN MODEL.

Setting	Value
Model	Single-layer 1D-CNN
Optimizer	Adam
Learning rate	0.001
Loss function	Mean squared error (MSE)
Batch size	32
Epochs	10

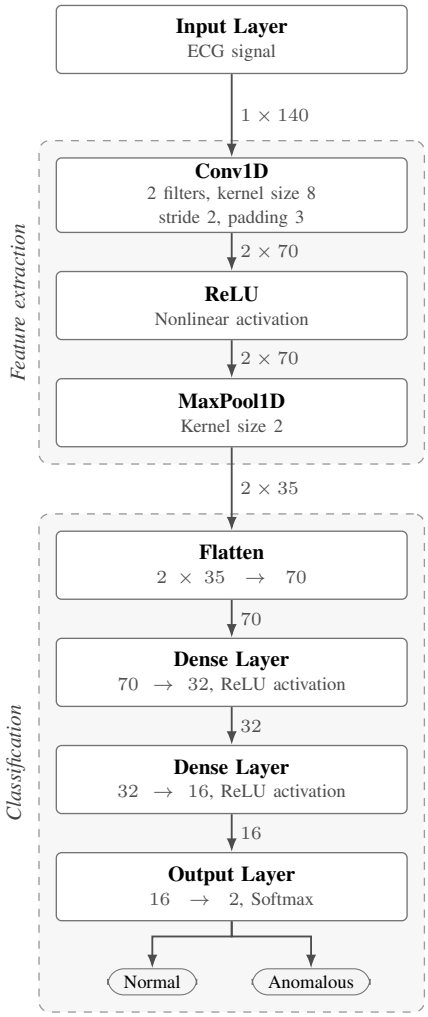


Fig. 5. Architecture of the 1D-CNN model used for ECG classification. Values on the arrows show the tensor shape after each layer.

The training and validation MSE loss of the model over 10 epochs is given in Fig. 6. Both losses decrease rapidly during the first two to three epochs: the training loss falls from roughly 0.22 in the first epoch to about 0.02 by the third epoch, while the validation loss drops from about 0.15 to below 0.03 over the same period. After this point, the improvement slows down and both curves gradually flatten, ending at approximately 0.01–0.02 at epoch 10. Since the training and validation curves remain close to each other and the validation loss does not increase, there is no clear sign of overfitting, and the model appears to generalize well to unseen heartbeats. The validation loss stays slightly above the training loss after the third epoch, which is the expected behavior.

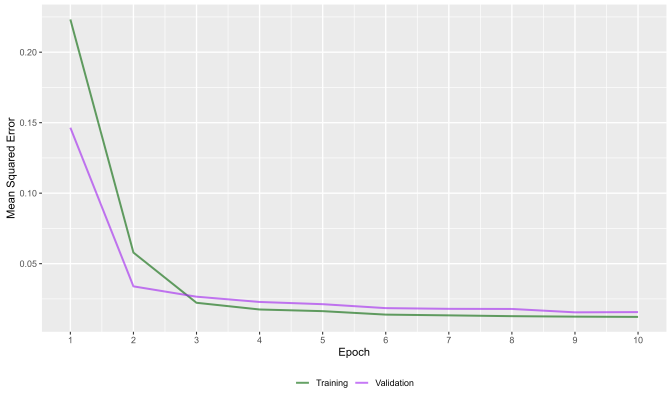


Fig. 6. Training and validation MSE loss of the 1D-CNN model over 10 epochs.

In the literature, several studies have adopted SMOTE to mitigate class imbalance and have employed Logistic Regression (LR) and Support Vector Machine (SVM)-based approaches. Also, Explainable Artificial Intelligence (XAI) techniques, particularly SHAP, have been used to enhance the interpretability of model predictions (Nayana and Vinay, 2026).

VI. LSTM Autoencoder for Anomaly Detection

The second part of the study considers a reconstruction-based approach. A Long Short-Term Memory (LSTM) network processes a sequence step by step and keeps a memory cell c_t that is controlled by three gates (Aggarwal, 2023). At time step t, with input x_t and previous hidden state h_{t-1}, the forget gate f_t, the input gate i_t and the output gate o_t are computed as

$$f_t = \sigma(W_f[h_{t-1}, x_t] + b_f)\,, \tag{5}$$
$$i_t = \sigma(W_i[h_{t-1}, x_t] + b_i)\,, \tag{6}$$
$$o_t = \sigma(W_o[h_{t-1}, x_t] + b_o)\,, \tag{7}$$

and the memory cell and hidden state are updated as

$$\tilde{c}_t = \tanh(W_c[h_{t-1}, x_t] + b_c)\,, \tag{8}$$
$$c_t = f_t \odot c_{t-1} + i_t \odot \tilde{c}_t, \tag{9}$$
$$h_t = o_t \odot \tanh(c_t), \tag{10}$$

where σ is the sigmoid function and $\odot$ denotes element-wise multiplication. The gates allow the network to keep or forget information over many time steps, which makes it suitable for modelling both short-term and long-term dependencies in ECG signals.

An LSTM autoencoder consists of an encoder that compresses the input heartbeat $\mathbf{x}$ into a low-dimensional representation $\mathbf{z}$ and a decoder that reconstructs the signal $\hat{\mathbf{x}}$ from $\mathbf{z}$. When the model is trained only on normal heartbeats, it learns to reconstruct normal patterns well, whereas anomalous heartbeats, which have a different shape, are reconstructed

poorly. The reconstruction error of a heartbeat of length $T = 140$ is

$$E(\mathbf{x}) = \frac{1}{T} \sum_{t=1}^{T} (x_t - \hat{x}_t)^2 , \tag{11}$$

and a heartbeat is flagged as anomalous if $E(\mathbf{x}) > \tau$, where τ is a threshold. The threshold is typically selected from the distribution of reconstruction errors on normal data (Shi and Jiang, 2026). Since the model needs no labelled anomalies during training, this approach is well suited to healthcare settings, where abnormal examples are scarce and costly to annotate (Shi and Jiang, 2026; Alamr and Artoli, 2023).

VII. Evaluation Metrics

The performance of the models is evaluated with metrics derived from the confusion matrix, where the anomalous class is treated as the positive class. Let TP, TN, FP and FN denote the numbers of true positives, true negatives, false positives and false negatives. The metrics are defined as

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}, \tag{12}$$

$$\text{Precision} = \frac{TP}{TP + FP}, \tag{13}$$

$$\text{Recall} = \frac{TP}{TP + FN}, \tag{14}$$

$$\text{Specificity} = \frac{TN}{TN + FP}, \tag{15}$$

$$F_1 = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}. \tag{16}$$

In a healthcare setting, recall is particularly important, because a missed anomalous heartbeat (a false negative) is usually more costly than a false alarm. Since the classes are imbalanced, accuracy alone can be misleading, and precision, recall and the F_1-score are therefore reported together with the area under the ROC curve (AUC), which measures how well the model separates the two classes over all possible thresholds.

VIII. Explainable Artificial Intelligence (XAI)

Deep learning models are often considered black boxes, which is a limitation in healthcare, where clinicians need to understand why a heartbeat is flagged as abnormal. Explainable artificial intelligence (XAI) techniques, particularly SHAP (SHapley Additive exPlanations), are therefore widely used to improve the interpretability of model predictions. Since the ECG5000 data are time-series data, the SHAP analysis in this study is planned to be performed over the time dimension, so that the time points that contribute most to the model's decision can be identified.

SHAP is based on the Shapley values of cooperative game theory. For a model f and a set of input features F, the contribution of feature i to the prediction for a given input is

$$\phi_i = \sum_{S \subseteq F \setminus \{i\}} \frac{|S|!\,(|F| - |S| - 1)!}{|F|!} \left[f(S \cup \{i\}) - f(S) \right], \tag{17}$$

and the contributions satisfy $f(x) = \phi_0 + \sum_i \phi_i$, where ϕ_0 is the average model output. For the ECG5000 data, the 140 time points of a heartbeat are the features, so each time point receives a contribution value. Plotting these values along the time axis shows which parts of the heartbeat, for example the region around the peak at time point 105 of the normal waveform in Fig. 4, push the model towards the normal or the anomalous class (Nayana and Vinay, 2026). Such explanations help to check whether the model relies on physiologically meaningful segments of the signal rather than on artefacts.

IX. Conclusion and Future Work

This study examined anomaly detection in healthcare applications using the ECG5000 heartbeat dataset. The exploratory analysis showed that the dataset is dominated by the Normal and R-on-T PVC classes, that the rarest classes (PVC, SP and UB) contain very few samples, and that the mean waveform of the Normal class differs clearly from those of the anomalous classes. A single-layer 1D-CNN was then trained on the raw 140-point signals, and its training and validation losses decreased together over 10 epochs without any clear sign of overfitting. The related studies reviewed above show that LSTM-based and attention-based reconstruction models are strong alternatives for the same task.

This study has some limitations. All experiments and visualizations were implemented in the R programming language, which restricted the range of deep learning architectures and explainability tools that could be explored within the scope of this work. The results should therefore be seen as a first step, and implementing the same models in other deep learning frameworks would make it possible to test a wider set of architectures and to compare the findings.

The next steps of this work are to train the LSTM autoencoder on the normal heartbeats and to select the anomaly threshold from the reconstruction error, to evaluate both models with the metrics given above, and to compare the results with the studies in Table I. The class imbalance can be further addressed with techniques such as SMOTE or class weighting, and SHAP will be used over the time dimension to explain which parts of the heartbeat drive the predictions. Finally, validating the models on a second dataset such as MIT-BIH Arrhythmia would show how well the findings generalize. This is important because the generalization of models across datasets and diverse patient populations is still described as an open challenge (Adegoke-Elijah et al., 2025; Wu et al., 2025), and Alamr and Artoli, who validate the same model on both ECG5000 and MIT-BIH Arrhythmia, report a clearly lower accuracy on MIT-BIH Arrhythmia (89.5%) than on ECG5000 (99%) (Alamr and Artoli, 2023).

References

A. Adegoke-Elijah, T. O. Ojewumi, and K. O. Jimoh, "ECG anomaly detection: a deep learning perspective with LSTM encoders," *African Scientific Reports*, vol. 4, art. no. 359, 2025. [Online]. Available: https://doi.org/10.46481/asr.2025.4.3.359

S. Aggarwal, “LSTM based anomaly detection in time series for United States exports and imports,” MPRA Paper No. 117149, Munich Personal RePEc Archive, Apr. 2023. [Online]. Available: https://mpra.ub.uni-muenchen.de/117149/

A. Alamr and A. Artoli, “Unsupervised transformer-based anomaly detection in ECG signals,” *Algorithms*, vol. 16, no. 3, p. 152, 2023. [Online]. Available: https://doi.org/10.3390/a16030152

L. Ferreira, “ECG5000 Dataset,” *OpenML*, 2022. [Online]. Available: https://www.openml.org/d/44794

A. Gharehbaghi, *Deep Learning in Time Series Analysis*. CRC Press, 2023. [Online]. Available: https://doi.org/10.1201/9780429321252

K. Nayana and S. Vinay, “Explainable AI-powered ECG anomaly detection using SHAP and LSTM autoencoders,” *Engineering, Technology & Applied Science Research*, vol. 16, no. 3, pp. 34998–35005, 2026. [Online]. Available: https://doi.org/10.48084/etasr.17734

S. Paul and S. Mishra, “LAC: LSTM autoencoder with community for insider threat detection,” in *Proc. 4th Int. Conf. Big Data Research (ICBDR)*, Tokyo, Japan, Nov. 2020. ACM. [Online]. Available: https://doi.org/10.1145/3445945.3445958

I. H. Sarker, “Machine learning: Algorithms, real-world applications and research directions,” *SN Computer Science*, vol. 2, no. 3, p. 160, 2021.

H. Shi and Z. Jiang, “LAVAE-Net: A deep LSTM-attention VAE framework for detecting ECG anomalies,” *Biomedical Signal Processing and Control*, vol. 118, p. 109755, 2026. [Online]. Available: https://doi.org/10.1016/j.bspc.2026.109755

L. Wu, T. Li, M. K. Majahar Ali, C. Ni, Y. Tian, and X. Zhou, “Adaptive Toeplitz convolution-enhanced classifier for anomaly detection in ECG big data,” *Journal of Big Data*, vol. 12, art. no. 56, 2025. [Online]. Available: https://doi.org/10.1186/s40537-025-01113-w